Jenkins Pipeline for MyGradleApp

Step 1: Open the Jenkins Dashboard in order to create a new item.

Step 2: Click on New Item in order to create a New Item. Give the name as "MyGradleApp" and Item type as Pipeline, then click Create.

Step 3: Configure the settings by defining the Definition as "Pipeline Script from SCM" and SCM as "Git". Paste the Repository URL of the project from GitHub and choose Credentials as "GitHub Credentials" which was created earlier.

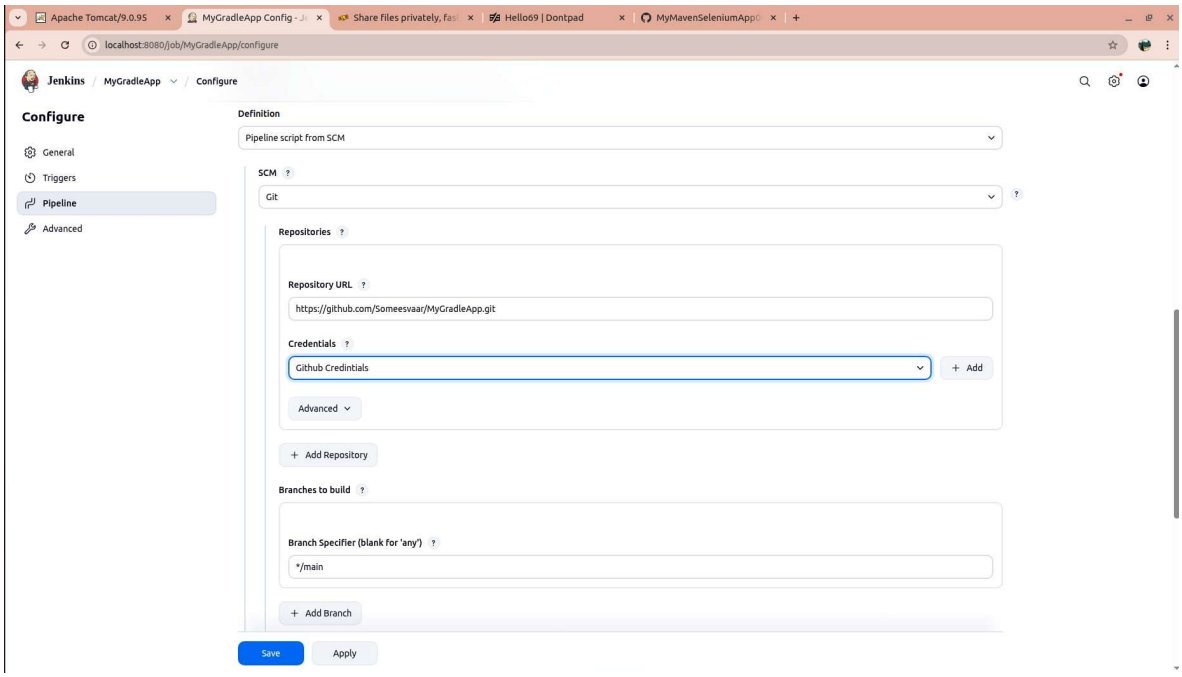

Fig F-1: Jenkins Pipeline configuration – Repository URL for MyGradleApp.git with GitHub Credentials

Step 4: In Terminal, go to the directory created for MyGradleApp and create a new file called "Jenkinsfile".

Step 5: Paste the following code inside the Jenkinsfile:

```
pipeline {
    agent any  // Use any available agent

    tools {
        gradle 'Gradle'  // Ensure this matches the name configured in Jenkins
        jdk 'JDK'
    }
    stages {
        stage('Checkout') {
            steps {
                git branch: 'main', url: 'https://github.com/Someesvaar/MyGradleApp.git'
            }
        }

        stage('Build') {
            steps {
                sh './gradlew build'  // Run Gradle build
            }
        }

        stage('Test') {
```

```
            steps {
                sh './gradlew test'  // Run unit tests
            }
        }

        stage('Run Application') {
            steps {
                // Start the JAR application
                sh 'java -jar build/libs/MyGradleApp-1.0-SNAPSHOT.jar'
            }
        }

    }
     post {
        success {
            echo 'Build and deployment successful!'
        }
        failure {
            echo 'Build failed!'
       }
    }
}
```

Step 6: Update pom.xml to change the versions of source and target from 1.7 to 1.8.

Step 7: Push the changes and commit by pushing the "Jenkinsfile" to the respective repository on GitHub.

Step 8: Go to Jenkins and Build the Pipeline, then check the Stage View to confirm successful execution.

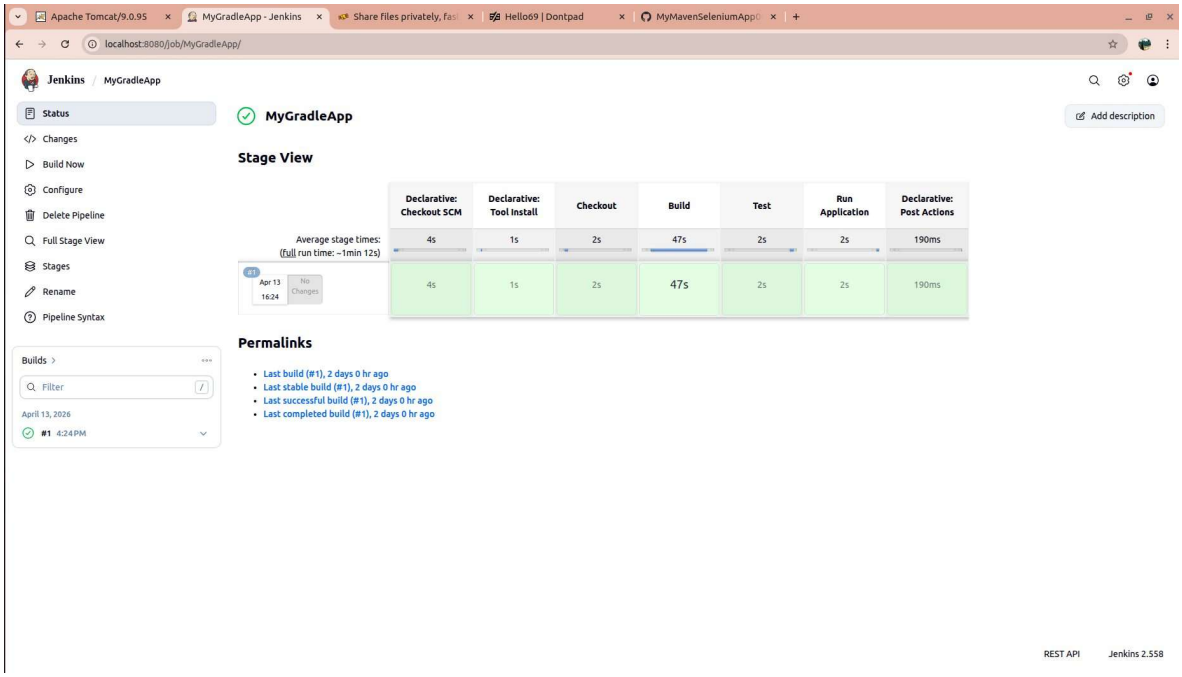

Fig F-2: Jenkins Stage View showing Build #1 successful for MyGradleApp (~1min 12s total)